Step 2: Then, let us find remaining letters to complete the code.

Plain Text	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Cipher Text	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V

f v i e o m r g s h i w g e r f i x v m g o c

B R E A K I N C O D E S C A N B E T R I C K Y

Thus, the secret sentence is decoded as, **BREAK IN CODES CAN BE TRICKY**

2. Substituting Cypher Text

Each letter in a text is "substituted" by certain pictures and symbols, a key message, group of words, letters or a combination of these can be used to encode or decode the information. Let us learn more from the following situation.

Situation:

The teacher divides the class into two teams and displays a worksheet given in the figure below for students to complete. Play a game, teams take turns suggesting letter substitutions and someone entering your suggestions in the table. Teams earn points for correct letter guesses.

Worksheet 1 :- Additive Cipher [key = 5]

Convert a given plain text into Additive Cipher text (number) code.

" mathematics is a unique symbolic language in which the whole world works and acts accordingly."

Tips for cracking additive ciphers:

- Especially for additive Ciphers, you only need **key number**. You can complete the cipher table by filling in the rest of the numbers in order.
- If you can, find and make **frequency table for alphabets**, it is help you to get started.
- Match the most **repeated letters** first and fill the Cipher number.
- Look for familiar one-letter words like **a** or **i**. common two and three letter words like **of, to, in, it, at, the, and, for, you …..**
- Look for **consecutive numbers** in the Cipher text and match them with possible **consecutive letters** in the plain text.

Now, to start converting a plain text into Cipher number, first we have to make a cipher table as shown below. Here there **key is 5**. As per key number, we have to start and fill **a = 05, b = 06 …….. z = 04** respectively.

Plain Text	**a**	**b**	**c**	**d**	**e**	**f**	**g**	**h**	**i**	**j**	**k**	**l**	**m**	**n**	**o**	**p**	**q**	**r**	**s**	**t**	**u**	**v**	**w**	**x**	**y**	**z**
Numbers	**00**	**01**	**02**	**03**	**04**	**05**	**06**	**07**	**08**	**09**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**
Cipher Numbers	**05**	**06**	**07**	**08**	**09**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**00**	**01**	**02**	**03**	**04**

To start encoding the text, let us count the frequency of alphabets and frame frequency table as shown below.

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